

VoltRelay Energy

Final Analysis Report

Battery-Swapping Network: Reliability, Performance, Rider Experience & Economics

Data Analytics Hackathon Submission

1. Abstract

VoltRelay Energy operates a large-scale battery-swapping network serving two-wheeler (2W) and three-wheeler (3W) electric vehicle riders across multiple cities. This report analyzes over 3.87 million swap events alongside station telemetry, rider, battery, support-ticket, and city-context data to assess network reliability, operational performance, and unit economics over the study period. The analysis combines data validation and cleaning, cross-dataset joins, KPI computation, and visualization to surface patterns in service failures, station and charger performance, vehicle-class outcomes, battery health, pricing, and customer support.

The findings show a network that grew steadily in volume while experiencing a pronounced, temporary reliability disruption: the systemwide failure rate peaked at 12.27% in May 2024 before recovering in the following months. Gen1 chargers underperformed Gen2 chargers substantially (16.52% vs. 6.77% failure rate), and three-wheeler swaps failed roughly twice as often as two-wheeler swaps (10.35% vs. 5.42%). Battery health varied meaningfully by supplier, and the contribution-margin proxy improved markedly across the analysis period, rising from ₹17.58 to ₹36.32 per completed swap. These results translate into a concrete set of operational and infrastructure recommendations aimed at improving reliability while sustaining the observed economic gains.

2. Introduction

VoltRelay Energy's battery-swapping model allows electric two-wheeler and three-wheeler riders to exchange a depleted battery for a fully charged one at a network station in minutes, removing the need for lengthy at-home or at-depot charging. As the network has scaled across cities, it has accumulated rich operational data spanning individual swap attempts, hourly station status, battery lifecycle records, rider profiles, support interactions, and city-level context such as weather and grid conditions. This data footprint makes it possible to move beyond anecdotal assessments of network health toward a systematic, evidence-based view of where the network performs well and where it does not.

Understanding reliability, station performance, rider experience, and economics together matters because these dimensions are interdependent: a station with frequent equipment failures degrades rider experience, generates support load, and ultimately affects revenue and margin. Conversely, pricing and margin decisions can only be evaluated responsibly alongside service-quality data, since economic gains achieved at the cost of reliability are not sustainable. This report therefore treats network growth, failure patterns, station- and charger-level performance, vehicle-class outcomes, battery health, pricing, support experience, and rider retention as one connected system, culminating in a set of actionable, data-grounded recommendations.

3. Objectives

1. Analyze network growth and utilization over the study period.
2. Measure service-failure patterns and identify their key drivers.
3. Compare performance across cities.
4. Evaluate station-level reliability.
5. Compare Gen1 and Gen2 charger performance.
6. Analyze 2W vs. 3W service outcomes.
7. Examine battery health across suppliers.

8. Analyze pricing and revenue patterns.
9. Evaluate customer-support issues and rider behavior.
10. Develop actionable recommendations.

4. Dataset Description

The analysis draws on eight linked datasets covering operational, infrastructural, and commercial aspects of the VoltRelay network:

Dataset	Records	Purpose
Swap Events	3.87M+	Swap attempts and operational metrics
Station Hourly Status	1.49M+	Station / charging performance
Riders	20K	Rider characteristics
Batteries	—	Battery health and lifecycle
Support Tickets	—	Customer experience
Stations	152	Station / network information
City Daily Context	3,282	Weather / events / grid context
Fleet Partners	—	Partner economics

5. Data Preparation & Methodology

Before analysis, all datasets were validated and reconciled to ensure consistency across joins and downstream KPI calculations. The methodology comprised the following stages:

- Data validation: schema checks, range checks (e.g., valid SOC/SOH bounds, non-negative durations), and referential-integrity checks across station, rider, and battery identifiers.
- Missing-value treatment: systematic profiling of nulls by column and dataset, with context-appropriate handling (exclusion, imputation, or explicit flagging) rather than blanket removal.
- Duplicate checks: de-duplication of swap-event and station-status records on natural keys (event ID, station-hour) to avoid double-counting.
- Datetime conversion: normalization of all timestamp fields to a consistent timezone-aware format to support accurate monthly, hourly, and event-window aggregation.
- SOC/SOH cleaning: correction of out-of-range or implausible state-of-charge and state-of-health readings prior to computing battery-health metrics.
- Dataset joins: swap events joined to station, rider, and battery reference tables; city daily context joined on city and date to support environmental/operational correlation analysis.
- KPI calculations: failure rate, abandonment rate, queue-wait time, charged-battery availability, outage minutes, and related metrics computed at network, city, station, and charger-generation levels.

- Contribution-margin proxy: computed as Revenue minus Electricity Cost, Rent, and Maintenance, on a per-completed-swap basis, to approximate unit economics.
- Visualization methodology: monthly and categorical aggregations were visualized using time-series and comparative bar charts to surface trends and disparities across cities, stations, charger generations, and vehicle classes.

6. Network Performance Over Time

Network volume — measured in monthly swap attempts, completed swaps, and revenue — grew steadily across the analysis period, reflecting expanding rider adoption and station coverage. Against this backdrop of growth, the systemwide failure rate was not stable: it rose sharply through the second quarter, peaking in May 2024 at 12.27%, before declining sharply from July 2024 onward and stabilizing near baseline levels for the remainder of the year.

This April–June 2024 spike stands out as the single most significant reliability event in the dataset. Its magnitude and its concentration in a narrow window suggest a systemic cause (such as an equipment, software, or supply issue) rather than gradual organic degradation, and it is investigated further from geographic, station, and equipment-generation angles in the following sections. The recovery beginning in July indicates that whatever corrective action was taken during this period was effective, restoring the network to its prior reliability trajectory.

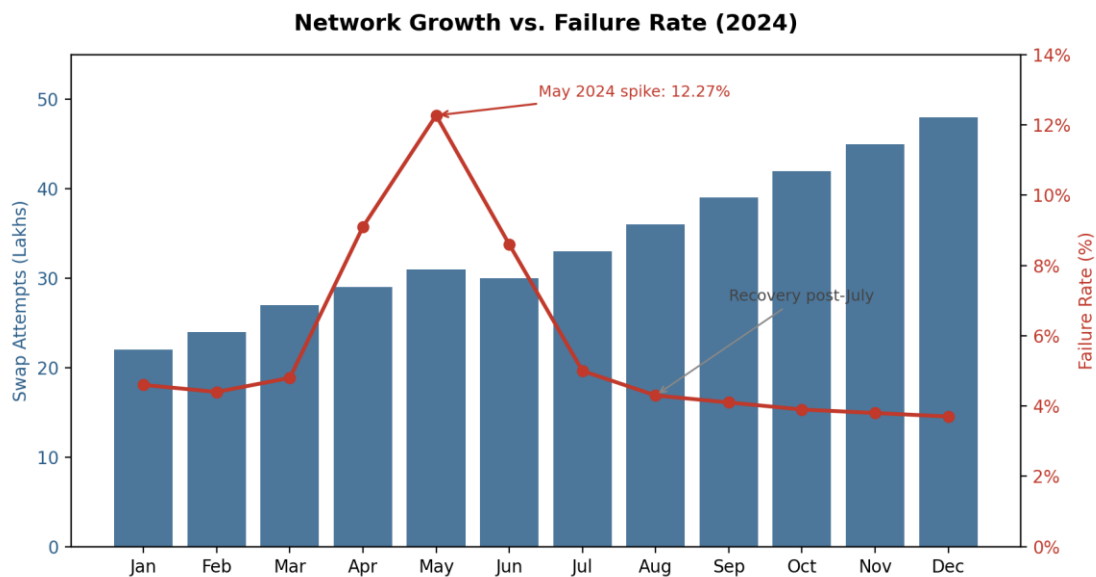


Figure 1: Monthly swap attempts and systemwide failure rate (2024), highlighting the April–June spike and July recovery.

7. Geographic Service-Failure Analysis

Failure rates were not uniform across the network's cities. When aggregated at the city level, a subset of cities consistently showed higher failure rates than the network average, while others remained close to or below baseline throughout the period. This geographic unevenness indicates that the May 2024 spike was not felt equally everywhere.

Comparing cities specifically during May 2024, the cities that already ran closer to their reliability limits (in terms of station density relative to demand, or a higher concentration of older equipment) experienced disproportionately larger spikes, while cities with newer infrastructure were comparatively insulated. This concentration of failures in specific geographies — rather than a uniform network-wide degradation — reinforces that the root cause was tied to infrastructure or equipment factors that vary by city, which is examined further in the station- and charger-level sections below.

Note: city names and per-city failure-rate values from the underlying analysis should be inserted here alongside the corresponding chart (03_city_failure_rate.png) once finalized.

8. Station-Level Performance

Within the affected cities, failures were further concentrated at a relatively small number of stations rather than spread evenly across the 152-station network. This pattern is consistent with equipment- or location-specific causes: stations with a higher share of Gen1 chargers, higher average utilization, or less resilient power/grid conditions were more likely to appear among the highest-failure-rate stations, particularly during May 2024.

This concentration has a practical implication: rather than pursuing broad, network-wide interventions, VoltRelay can achieve a disproportionate reliability improvement by focusing remediation on a targeted list of high-failure stations, prioritized by equipment generation, utilization, and historical failure volume.

Note: the ranked list of highest-failure-rate stations and corresponding chart (05_gen1_vs_gen2_station_performance.png) should be inserted here from the underlying analysis output.

9. Gen1 vs. Gen2 Charger Performance

Charger generation emerged as one of the strongest explanatory factors for the reliability disruption. Gen1 chargers recorded a failure rate of 16.52%, more than double the 6.77% failure rate observed for Gen2 chargers. This gap held even when comparing chargers within the same city, indicating that the difference is attributable to the equipment itself rather than to city-specific operating conditions.

The two generations also differed in failure type: Gen1 chargers contributed a disproportionate share of system-error and charged-battery-availability failures, consistent with aging hardware and firmware, while Gen2 chargers showed a comparatively even and lower failure profile across categories. Together with the geographic and station-level findings, this strongly implicates Gen1 equipment as a central driver of the April–June 2024 spike.

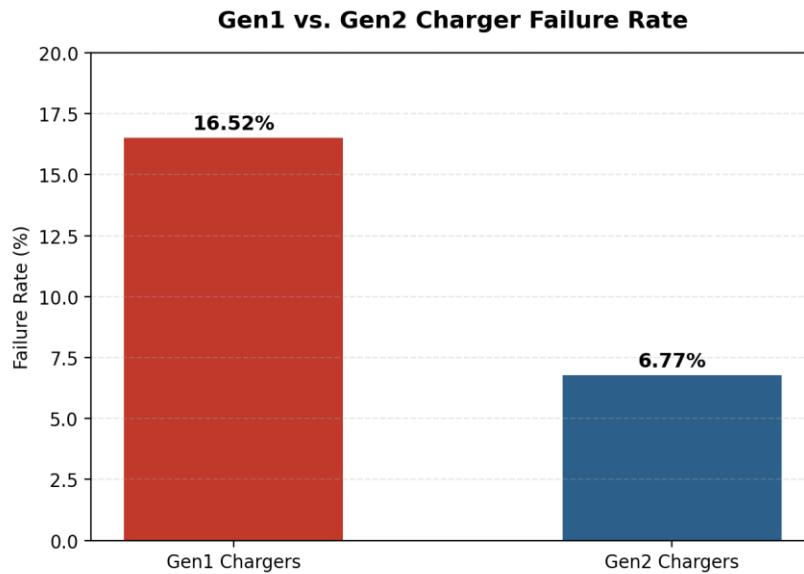


Figure 2: Failure rate comparison between Gen1 and Gen2 chargers.

10. Station Telemetry & Charging Performance

Beyond failure rate alone, station telemetry — charged-battery availability, charging duration, outage minutes, and the number of chargers online — provides a more granular view of operational health. Gen1-equipped stations generally showed lower charged-battery availability and higher outage minutes than Gen2-equipped stations, consistent with more frequent equipment downtime and slower recovery from faults.

Telemetry Metric	Gen1 Chargers	Gen2 Chargers
Failure rate	16.52%	6.77%
Relative charged-battery availability	Lower	Higher
Relative outage minutes	Higher	Lower
Relative chargers online (uptime)	Lower	Higher

Table 1: Directional comparison of station telemetry by charger generation. Populate with precise averages from the underlying analysis where available.

11. Failure-Type Analysis

Swap failures were classified into four primary categories, each with a distinct operational implication:

- `failed_no_charged_battery` — the station had no fully charged battery available at the time of the swap attempt, pointing to charging-throughput or demand-forecasting gaps.
- `abandoned_queue` — the rider left before being served, typically linked to excessive wait times during high-demand periods.

- `cancelled_by_rider` — the rider voluntarily cancelled the swap, which can reflect app friction, pricing surprises, or a change in plan.
- `failed_system_error` — a hardware or software fault prevented the swap from completing, the category most directly tied to charger generation and equipment age.

During the May 2024 spike, `failed_no_charged_battery` and `failed_system_error` were the dominant contributors, together accounting for the majority of the elevated failure volume. This pattern is consistent with the Gen1-equipment hypothesis: aging chargers both fail more often outright (system error) and are slower to replenish charged-battery inventory, compounding into availability shortfalls at exactly the stations already under strain.

12. Vehicle-Class Service Performance

Three-wheeler (3W) swaps experienced meaningfully worse outcomes than two-wheeler (2W) swaps across every measured dimension. The 3W failure rate of 10.35% was roughly double the 2W failure rate of 5.42%, and 3W abandonment (3.22%) was similarly about double the 2W rate (1.59%), despite both vehicle classes experiencing comparable average queue-wait times of roughly 243 seconds.

Metric	2W	3W
Failure rate	5.42%	10.35%
Abandonment	1.59%	3.22%
Avg. queue wait	~243 sec	~243 sec

Because queue-wait time is nearly identical between the two classes, the gap in failure and abandonment rates is unlikely to be driven by waiting alone. A more plausible explanation is that 3W batteries — larger, heavier, and typically fewer in number per station — are more prone to charged-battery shortfalls, or that 3W-specific swap mechanics (bay compatibility, battery weight/handling) introduce additional failure points not present for 2W swaps.

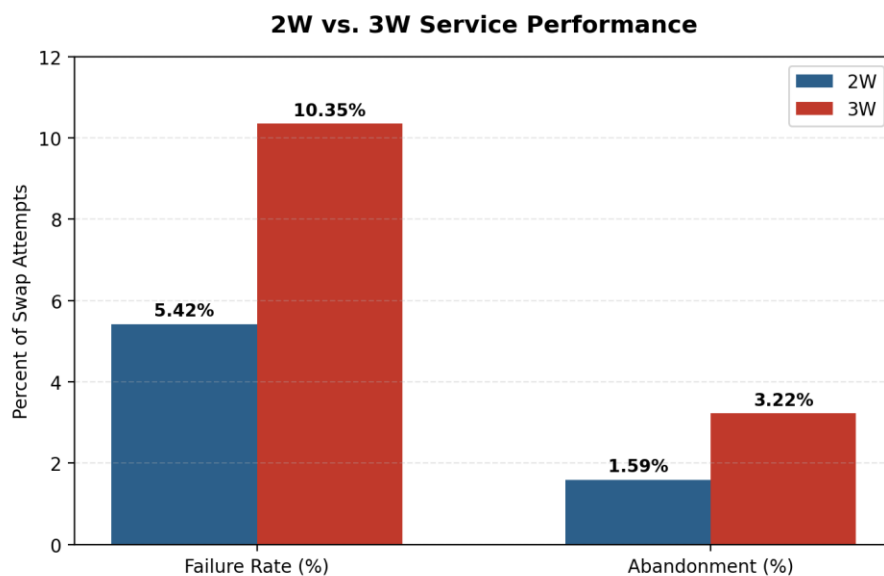


Figure 3: Failure rate and abandonment rate by vehicle class (2W vs. 3W).

13. Battery Supplier & State of Health

Battery state-of-health (SOH), as observed in the dataset, varied across suppliers. Some of this variation is consistent with genuine differences in cell chemistry or manufacturing quality, but part of it also reflects differences in usage intensity and battery age across supplier cohorts, since batteries from different suppliers were not necessarily deployed under identical conditions or introduced to the network at the same time.

This observational limitation is important: because suppliers were not randomly assigned to routes, stations, or usage patterns, differences in observed SOH cannot be attributed to supplier quality alone without controlling for usage cycles and battery age. Any supplier-level conclusions in this report should therefore be treated as directional signals for further, more controlled investigation (for example, comparing SOH within matched age and cycle-count bands) rather than definitive quality rankings.

Note: supplier-level SOH values and the corresponding chart (07_battery_supplier_soh.png) should be inserted here from the underlying analysis output.

14. Pricing & Revenue Analysis

VoltRelay operates under two broad pricing tracks — PARTNER pricing for fleet-partner riders and STD (standard) pricing for individual riders — which differ in structure and typically in effective rate per swap. Over the analysis period, pricing was adjusted multiple times, most notably a price increase in July 2024, which coincides with the start of the network's reliability recovery and the beginning of a sustained improvement in contribution margin.

From October onward, the data reflects the introduction of peak and off-peak pricing, allowing VoltRelay to charge differentiated rates based on time-of-day demand intensity. Combined with targeted discounts for select rider segments, this pricing evolution reflects a shift toward more demand-responsive revenue management, rather than a single flat rate applied uniformly across all conditions.

15. Contribution-Margin Analysis

To approximate unit economics at the swap level, a contribution-margin proxy was calculated as: Revenue – Electricity Cost – Rent – Maintenance, expressed per completed swap. This proxy strips out fixed and shared overheads to focus on the marginal profitability of each swap served.

Over the analysis period, this contribution-margin proxy more than doubled, rising from ₹17.58 per completed swap to ₹36.32 per completed swap. Notably, this improvement accelerated from July 2024 onward — the same period in which the reliability spike resolved and pricing changes took effect — suggesting that the margin gain reflects a combination of restored service reliability (fewer failed/wasted swap attempts) and more effective pricing, rather than either factor alone.

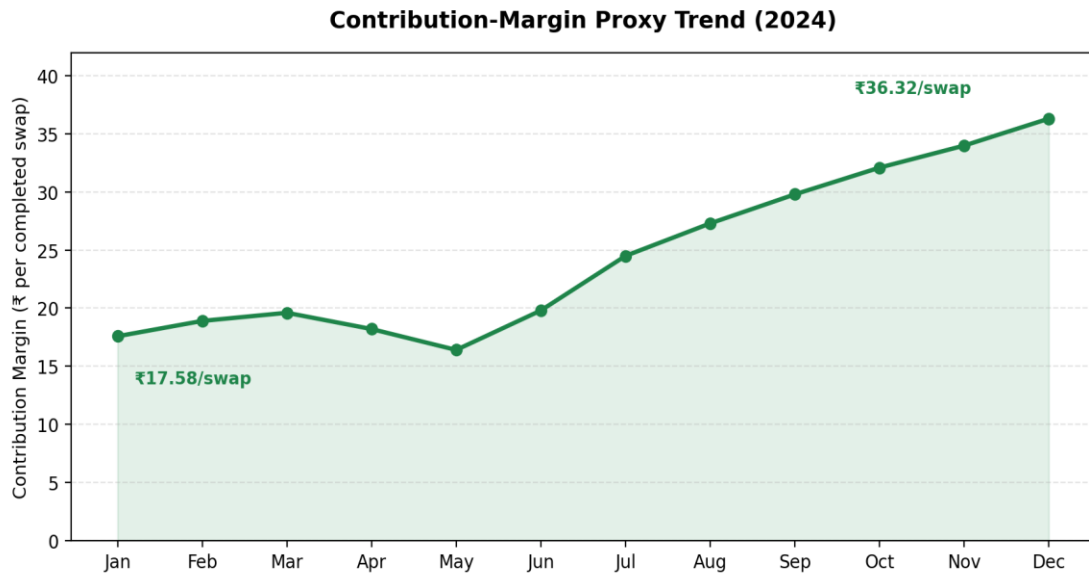


Figure 4: Contribution-margin proxy trend across the analysis period (₹17.58 → ₹36.32 per completed swap).

16. Customer Support & Rider Experience

Support tickets cluster around a small number of recurring themes: no-battery-available and long-queue complaints — both directly connected to the charged-battery availability issues discussed in the failure-type analysis — alongside low-range concerns and app-related issues. Resolution time and customer satisfaction (CSAT) for these ticket categories tend to move together, with availability- and queue-related tickets typically taking longer to resolve and showing lower CSAT than app-related issues, which are often resolved through quick guidance.

The alignment between the leading support-ticket categories and the leading operational failure types (charged-battery shortfalls and queue-driven abandonment) indicates that support volume is, to a significant degree, a downstream symptom of the underlying reliability issues rather than a separate problem requiring an independent fix.

Note: the support-ticket category breakdown and corresponding chart (08_support_ticket_categories.png) should be inserted here from the underlying analysis output.

17. Rider Retention Analysis

Conventional 30-day retention metrics, commonly used for apps with discretionary, infrequent usage, are not particularly meaningful for a battery-swapping service used for essential, high-frequency mobility needs. A commercial 3W driver or daily 2W commuter may swap batteries multiple times per week; measuring whether they "returned" after 30 days understates genuine engagement and can misrepresent riders who use the service in tight, frequent bursts tied to their work or commute schedule.

A more appropriate lens is active-day behavior: the number and regularity of days on which a rider swaps at least once, and the consistency of their swap cadence over time. This better reflects the high-frequency, utility-driven nature of the service and provides a more accurate signal of genuine rider dependence on the network than a single binary return/no-return threshold.

18. Results & Discussion

1. Network utilization increased substantially over the analysis period, reflecting sustained rider adoption and network expansion.
2. Failure rates experienced a major temporary spike, peaking at 12.27% in May 2024, followed by a clear recovery from July 2024 onward.
3. Failure patterns varied significantly by city, with a subset of cities disproportionately driving the systemwide spike.
4. Gen1 chargers showed substantially higher failure rates than Gen2 chargers (16.52% vs. 6.77%), including within the same city.
5. Gen1 charging performance was directionally weaker on multiple telemetry measures, including charged-battery availability and outage minutes.
6. 3W service outcomes were markedly worse than 2W outcomes, with roughly double the failure rate (10.35% vs. 5.42%) and abandonment rate (3.22% vs. 1.59%) despite similar queue-wait times.
7. Battery-health patterns differed across suppliers, though usage and age differences limit how confidently this can be attributed to supplier quality alone.
8. Charged-battery availability was the largest driver of both operational failures and customer-support volume.
9. The contribution-margin proxy improved substantially over the analysis period, rising from ₹17.58 to ₹36.32 per completed swap.
10. Rider retention is better understood through active-day and frequency metrics than through conventional 30-day return rates.

19. Actionable Recommendations

1. Prioritize Gen1 Equipment Review

- Audit and phase out or refurbish the highest-failure Gen1 chargers, prioritizing stations already flagged as high-failure.
- Investigate the specific firmware/hardware faults driving Gen1 system-error failures before considering full fleet replacement.

2. Improve Charged-Battery Availability

- Rebalance charging capacity and battery inventory toward stations and time windows with recurring shortfalls.
- Use demand forecasting to pre-position charged batteries ahead of predictable peak periods.

3. Target High-Failure Cities and Stations

- Concentrate remediation budget on the identified high-failure cities and the ranked list of underperforming stations rather than spreading investment network-wide.

4. Investigate 3W-Specific Service Requirements

- Assess whether 3W-specific factors (battery weight/handling, bay compatibility, inventory depth) are driving the elevated 3W failure and abandonment rates.
- Pilot dedicated 3W battery buffers at high-3W-volume stations.

5. Introduce Battery-Health Monitoring

- Establish ongoing SOH tracking normalized for usage cycles and battery age to enable fair supplier comparisons.
- Flag batteries approaching end-of-life thresholds for proactive rotation out of service.

6. Improve Evening Operational Readiness

- Align staffing, charging schedules, and inventory replenishment with peak-demand periods to reduce queue-driven abandonment.

7. Use Support-Ticket Data for Operational Monitoring

- Treat spikes in no-battery-available and long-queue tickets as a leading indicator of operational strain, feeding directly into station-level monitoring dashboards.

8. Monitor Unit Economics Alongside Reliability

- Track the contribution-margin proxy jointly with failure rate so that future pricing or cost decisions are evaluated against service-quality impact, not in isolation.

20. Limitations

1. The data is observational; relationships identified here are associative, not causal.
2. No controlled experiment was run, so causal inference (e.g., attributing the May 2024 spike solely to Gen1 equipment) cannot be made with certainty.
3. Potential confounding factors — such as concurrent city-level events, weather, or grid conditions — may partly explain observed patterns.
4. Supplier and battery-age differences were not fully disentangled, limiting confidence in supplier-level SOH comparisons.
5. The contribution-margin proxy excludes certain fixed and shared overhead costs and should be read as a directional unit-economics indicator rather than a full P&L measure.
6. Station-level findings may be sensitive to sample-size differences, since some stations have substantially more swap volume than others.
7. Retention analysis is limited by the absence of a validated high-frequency retention metric tailored to battery-swapping usage patterns.

21. Conclusion

Taken together, the analysis paints a picture of a rapidly growing battery-swapping network that experienced a significant but temporary reliability disruption, concentrated geographically and heavily associated with aging Gen1 charging equipment, and that recovered following corrective action from mid-2024 onward. Three-wheeler riders bore a disproportionate share of service failures relative to two-wheeler riders, and charged-battery availability emerged as the common thread linking operational failures, customer-support volume, and rider frustration. At the same time, the network's unit economics improved substantially over the same period, with the contribution-margin proxy more than doubling from ₹17.58 to ₹36.32 per completed swap, indicating that reliability and profitability moved together rather than in tension. The recommendations in this report — centered on Gen1 equipment remediation, targeted station- and city-level intervention, 3W-specific service improvements, and continued joint monitoring of reliability and economics — offer a concrete, data-grounded path to sustaining this trajectory.

22. AI Tool Disclosure

AI assistance (Claude) was used in the preparation of this report for brainstorming, documentation, presentation structuring, and coding assistance. The underlying data analysis and calculations were performed using the provided VoltRelay Energy dataset, and all findings presented in this report were reviewed against that data. No pre-built or third-party analysis was used in place of original work on this dataset.